INTRODUCTION TO COMPUTER

Information and communication plays an important role in our day-to-day activities. This information and communication has been greatly enhanced by the advent of computers. Computers can now be found in banks, shops, and government offices and media houses. It has therefore become very important that a person becomes a knowledgeable and competent user of computer based

Technologies or Information and Communication Technology (ICT) to survive in this technological world.

What a Computer does:

There are <u>four</u> jobs that any computer is built to do. These are:

1. <u>Accept</u> information. You must put information into a computer. This information is of two types namely:
   I. A collection of words and numbers
   ii. A set of instructions, instructing the computer what to do.
2. <u>Store</u> the information. The information is stored in a device called a <u>memory</u> that holds the information for as long as long as you want it.
3. <u>Process</u> information. The stored information is then worked upon and turned into what the computer was instructed to do.
4. <u>Give out</u> the processed information. The results of the processed information are then given out. They are the results of the instructions to the computer.

DEFINITION:

What is a computer?

- A computer is an electronic machine which can receive data, manipulate (process) it and output information
  - A computer is an electronic device that can accept data input, store, process and output information according to specified instructions. A working computer is made up of the hardware and software

THE BASIC PHYSICALPARTS OF A COMPUTER

A desk top computer is made up of collection of components /parts that are inter-connected to function as a single unit and these parts may include

THE SYSTEM UNIT: This is the part that houses the brain/inside parts of a computer among them being the CPU central processing unit

MUSE: Is a pointing device that enables the user to process, manipulate or execute a command. It is used to control an arrow displayed on the screen. To execute a
Command/task, the user moves the mouse, which also in return moves the pointer on the screen in the same direction

THE MONITOR: A monitor or simply the screen is a TV like device used for displaying information.
It is called the monitor because it enables a user to monitor (see) what is going on in a computer.

THE KEY BOARD: Is a device that enables a user to enter data into the computer

Keyboard keys and their functions

Tab: This key can be used to move forward through options in a dialog box.

Caps Lock: Locks the keyboard in "Capitals" mode (only applies to Alpha keys).

Shift: The use of this key is to allow selection of capital letters when depressing the alphabet characters, or selecting the characters above other non-alpha keys.
Control Key (Ctrl) : Depressing the Ctrl key while clicking allows multiple selections. Common actions performed by the combination Ctrl and other keys.
Ctrl + A: Selects **all** items

Ctrl + B: add or remove **Bold** formatting

Ctrl + C: **Copy**, places the selected/highlighted copy on the clipboard.

Ctrl + F: Opens the **Find what:** dialog box.

Ctrl + I: Add or remove **Italic** formatting.

Ctrl + N: opens a **New Document,** In most publishing programs like Word,.

Ctrl + o: Open, brings up a browse dialog and allows you to select a file to open.

Ctrl + P: **Print**

Ctrl + S: **Save**

Ctrl + U: Add or remove **Underline** formatting.

Ctrl + V: Paste, inserts the copy on the clipboard into the area where your flashing cursor is positioned or the area you have selected/highlighted.
Ctrl + W: Close, will close the document currently open window.

Ctrl + X: Cut, removes the selected/highlighted copy and places it on the clipboard.

Space Bar: Insert a space between words**.**

Enter: Creates a new Paragraph. It is referred to as a Hard Return.

Backspace: deletes characters to the left of the insertion point while working with text.

Characteristics of Computers

For any device to be called a computer, it must manifest some or all of the following characteristics:

- Speed
- Accuracy
- Storage
- Diligence
- Artificial intelligence
- Automation
- Versatility

1. *Speed*: Computers are very fast in their operation in that their speed is measured in millions of instructions per second

2. *Accuracy*: Computers are accurate. This means that they hardly make any mistake. Computers are capable of detecting and correcting any mistakes made. However, if wrong data is fed into the computer, wrong information is also given; hence the saying *"Garbage In, Garbage Out" (GIGO).* This implies that the accuracy of the computer largely depends on the accuracy of the user.

3. *Storage*: For the computer to be able to work, it should have some form of work space where data is stored before it is processed or where information is stored before being output to particular devices. This storage space is called the *Memory*.

4. *Diligence*: Computers are diligent. This means that computers have ability to perform the same task over and over again for a long time without getting tired or bored. Unlike humans, computers do not complain.

5. *Artificial Intelligence*: Computers have artificial intelligence. This means that they can respond to commands or requests given to them and provide solutions. The computer's intelligence is said t be artificial because it is enabled by the *computer programs* installed in them.
   A computer program is a set of instructions written in a computer language.

6. *Automation*: This means that computers have the ability to work without human supervision.

   Computers do not need any supervision in order to perform programmed routines. In other words, they are automatic.

7. *Versatility*: This means that a computer has the ability to perform more than one task at the same time. For example, a computer should be able to play music while one is typing.
The ability of the computer to perform more than one task is called *Versatility* or *Multi-task*

Terminologies and Concepts

Computer professional – a person who has had formal education in the technical aspects of computers e.g computer programmer.
Computer competence – this is applying some skills using a computer to meet your information needs and improve your productivity.
User – some one without much technical knowledge in computers but uses it to produce information.

Computer Literacy – having the knowledge and understanding of computers and their uses

Data is unorganized facts e.g. a list of students in school, irrespective of their class, sex or age.

-a collection of raw facts, figures and symbols like numbers,words,imagws,videos and sound given to the computer during the input stage/phase
Information is organized facts or data that has been turned into more meaningful form e.g. a list of students in school according to their classes, sex and age.
Data Processing is the procedure of transforming data into desired output

*Information Processing is the p*rocess of transforming data into which can be used to make better decisions (pay cheques, pay slips)
Data processing systems include all people, procedures, devices used to produce desirable output. Information-process-system is useful information that can be used in decision making.

THE COMPUTER SYSTEM:

A computer system refers to the combination of components that make up a complete and functioning computer

Let's compare the computer to the human being. A human being has got a body and a soul. These two parts works hand in hard for the existence of a human being. Incase one of the two is not there, there will then be no operation.

For this case, the hardware is the body of the computer; the software is the soul of the computer. Now the human ware will be the person who would like to use the machine this person uses the software to communicate with the hardware and get what he wants

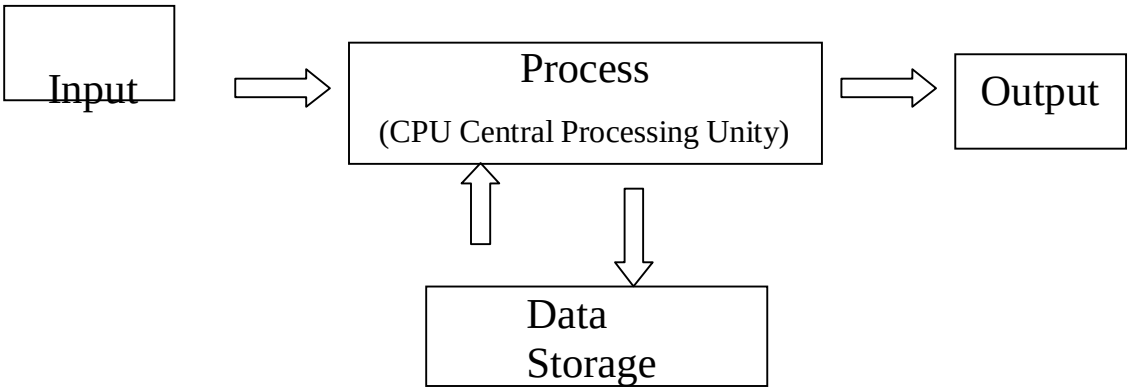
i) **Hardware** – These refer the physical and electronic components of the computer. E.g. Keyboard, Monitor, Mouse, etc.
ii) **Software** – These are electronic instructions that tell the computer how to perform a task. iii) **People / Human ware** – The trained personnel, who operate, design and develop computers

The data/information processing cycle: refers to the sequence of events that take place in the processing data to convert it into information. A computer to work there must be there basic things i.e.

- Human ware,
- hardware and
- Software.

The human ware

- The one who brings the hardware and software into interaction, in other word, human ware becomes the user of the computer. Someone who performs/ instructs the computer to perform tasks is the human ware. Remember computers need data to be inserted in order to produce results as information
- The data processing cycle starts with the human ware that generates the input facts.
- The human ware then sends his raw facts to the computer using any of the input devices.
- These facts will then go to the control unit processing and control Unit will forward it to the Arithmetic Control Unit to calculate the number of bits and bytes in order to determine the size of the data being worked on.
- The data will now be sent to the RAM for temporally storage (Primary Storage) and at the same time presented to the Visual Display Unit in kind of Information.
- Data will remain in the RAM until when the computer user decides to store it as information onto a secondary Storage Device.
- Diagram illustrating Data/information Processing



INPUT –it refers to entering data into the computer

Examples of computer parts used for input includes- keyboard, mouse, scanners, digital cameras joy sticks, trackball, touch screen, Microphones, etc.

PROCESSING-this refers to performing operations on the data.

The parts that used for processing is the CPU called the processor/micro processor brain
STORAGE –this refers to keeping data, instruction, and information in a location for future use. Examples used for storage RAM, hard disk, flash disk, CD, DVD etc
OUTPUT-this refers to presenting or displaying results (information) using output devices

Examples used for output monitors/screen, LCD projector, printers, speakers, Headsetsv

COMPUTER LABORATORY CARE AND MAINTENANCE

- This is a room where computer learning takes place and where computers and all its components are kept

HOW CAN YOU EASE EYESTRAIN WHEN WORKING AT THE COMPUTER

- Every 10 to 15 minutes, take an eye break
- Look into the distance and focus on an object for 20 to 30 seconds
- Loll your eyes in a computer circle
- Close your eyes and rest them at least one minute
- Blink your eyes every five seconds
- Place your display device about an arm's length away from your eyes with the
- Use large font
- If you wear glasses, ask your doctor about computer glasses
- Adjust the lighting.

GENERAL RULES AND SAFETY PRECAUTION WHEN USING A COMPUTER LAB

- The computer lab should be kept clean to avoid dirt affecting our computer
- Bags and pullovers should be left outside the computer lab to avoid loss of small computer components by the computer learners
- Since the lab is a study room, the learners should maintain silence and the room its self should be located in a quite environment
- The computer users should switch on and off computers in a proper way to avoid damages
- Food stuffs and drinks should not be brought in the computer lab
- The devices positions should be maintained and learners should avoid un necessary disconnections
- Computer lab and all other resources in it should be specifically for academic only.

- Playing music and watching pornography is not allowed
- Avoid smoking and exposing computer and its components to dust
- Proper shut down of computer should be followed to avoid disk and system failure so avoid abrupt switching off
- Handle delicate storage devices with care. Don't touch the inner surface of computer disk and always remove safely flash disks from the systems
- Always sit upright to avoid muscle pain and back aches caused by poor sitting posture
- Don't bring magnetic device to the lab. Because a computer has magnetic disk which can be spoilt if they come near other magnetic fields
- Un necessary movements are not allowed to prevent learners from stepping on wires and devices
- Avoid running in the computer lab but walk gently and confidently. Why. *As you run, wire may trap you and you fall and damage other hardware in the computer*
- *Always respect other peoples work you find in the computer so do not delete, edit or even try to open any.*
- Do not open metallic cover of computer or peripheral devices particularly when the computers power is on
- Keep the room at a minimum temperature
- Keep fire sources out and far from the computer lab
- Removable storage devices should be scanned before opening them in computers to avoid viruses

COMPUTER LABORATORY SECURITY

Since the computer lab is a place of learning ,it receives many people and this may result into possible dangers like:

- Fire outbreaks, loss of gadgets, electric damages and others
- Thus there is need to employ some of the following in our computer lab
- Employing security personel:to guard the room 24hrs
- Air conditioners: these can help to keep the room at a very good temperature that cannot affect people and the machines. here you can buy fans, increase on the ventilators, put a ceiling in the room or buy the modern air conditioner
- Always use surge protectors such as uninterruptible power supply (UPS) or voltage stabilizers to ensure steady power supply

- Voltage stabilizers: helps to regulate the level of voltage/electricity that enters computer and other devices in lab so that electricity does not affect our computer.
- Un interruptible power supply (UPS) This device can also backup power and this can help your devices to remain on for some minutes when power goes off
- Security cameras: you can install security cameras (closed circuit television -CCTV) which can help you inspect the place to avoid people that many need to move out with some gadgets
- Alarms : you can attach alarms to every device in the lab and if some touches on the device, the buzzer or alarm will make sound

IMPORTANCE OF SERVICING AND MAINTAINING ACOMPUTER

- It speeds up your computer
- It's a preventive measure to protect your computer from viruses and other malware
- Prevents loss of information-hackers ,intruders can unauthorized access information
- Increases productivity of both software and hardware-new vision of software and hardware performs better than older one
- Gives a computer long life-just like a person who cares a bout him self stay health and live along productive life
- Upgrading your pc will keep it up to date with technology and add extra features that your basic pc did not have upon purchase
- Prepares for faster crush recovery-backing up

BOOTING A COMPUTER

This is the process of starting or resetting a computer. The computer automatically goes through the process of self-test and preparation for use.

Types of Booting

i) ***Cold Booting*:** This starting of a computer, which has originally off. The computer checks all its components to determine whether they are working properly. This process is called POST (Power On Self Test).

ii) ***Warm Booting*:** This refers to restarting a computer, which has been originally on. It is forced to restart by pressing CTRL+ALT+DEL.

- Is the process of restarting a computer that is ready turned on by used of command or as a result of instruction
- You can restart by pressing CTRL+ALT+DEL keys together

- Sometimes the computer stops working or responding to the commands given by the user, in such cases we need not switch the power off ,we can restart the system by pressing reset switch
- When the computer freezes (Does not respond to any input Instruction by the mouse or the Keyboard).
- After changing CMOS(complementary metal-oxide semi conductor) settings AIT, control &Del – it remains
- When there is a faulty, (Computer Bypasses the 1st Steps of booting)
- After installing/ hardware
- After installing new software

INFORMATION COMMUNICATION TECHNOLOGY

- The technology that handles information as well as aiding communication.
- The integration of computers and telecommunication facilities for the purpose of communication
- **Information:** Processed Data. Something useful to a recipient that can be used for decision making. In other words it's the end result of data processing. E.g. CVs, Transcripts, Payrolls, Balance Sheets, Final %s of votes.
- **Communication:** The transmitting of a message from one point to another or a sender to a receiver through several (physical or electronic) media.
- **Technology:** The advanced scientific knowledge that eases work. However with ICT we concentrate on
- Therefore, it **refers** to the procedures and equipment used by people to improve the quality of gathering, processing, communicating and storing information.

USES AND FUNCTIONS OF COMPUTERS

Uses of Computers

Computers are widely used in our everyday life to accomplish large number of useful applications or various tasks.
Some of the areas that computers are widely used include:-

- Business.
- Military.
- Education.
- Research.
- Science and Engineering.
- Entertainment and Leisure (Recreation)

1. BUSINESS APPLICATIONS.

- They are used to carryout business data processing which includes designing receipts, bills, invoices etc
- They are used to carry out financial analysis which includes analyzing business performance, making sales summary, and forecasting business needs like labor
- They are used in vending machines, elevators, cars, and phones to automate them.

  These machines are automatic because of the special computers that control them
- They are used to reduce inventory costs by keeping all documents in electronic form instead of paper files which consume a lot of space
- They are used in banks to transfer money electronically which has eased the process of banking and the risks involved in transporting cash
- They are used in banks to work as tellers. These ATMs have lessened the congestion that was in banking halls

2. SCIENCE AND ENGINEERING:

Today, the computer is used and can help in almost every area of science and engineering.

- Computers can make millions of complex calculations per second this help in scientific communication which requires speed and accuracy in making complex computations.
- Help astronomers test physics theories. Needed in forecasting complicated weather patterns.
- They are used to unlock the mysteries of DNA molecules.

  Help in processing and enhancing images or pictures; a process called *image processing*.

3. **HEALTH CARE:**

- They are used to carry out medical research which has brought about new medicines
- They are used to develop colorful graphic scans that can show the affected parts of the body
- They are used to make the diagnosis of the disease and prescribe the medicine
- Computers are used in ambulances to locate patients and the scene of the accident
- Computers are used to help people lose weight by analyzing nutrition of a meal
- They are used by doctors to carry out medical collaboration whereby doctors work together to come up an appropriate remedy
- Computerized machines such as the CT Scan, Magnetic imaging and ultra sound devices have enabled the diagnosis and cure of many diseases
- They are used to demystify (test) DNA molecules and establish people's true parentage

4. **SPORTS:**

- Computers are used to analyze and design new plays, make draft picks and day to day business operation of any sports business
- They have also been used to analyze the smallest movements made by the athletes

- Computers have been used by experts to pin point wasted energy and movements for many athletic events. This brings about improvement in performance
- A special tennis computer called *computerennis,* records the important plays of a match. Then the data is used to determine how a player can improve.
- They are used to relay or broadcast sports live events like live matches on Sky sports

ADVANTAGES OF USING COMPUTERS

1. They have improved communication facilities and communication in society. This is seen in the usage of electronic mails, telephone exchange, social networks like facebook, voice calls through skype; where people can send and deliver messages instantly.
2. They have enabled the sharing of data and information between people in different areas. This has facilitated easy exchange of data and ideas and thus increased the speed for executing operations.
3. Computers have increased data accuracy, reliability and data integrity. This is because computers do not make mistakes because they have self-check systems which can detect errors which detect and correct errors
4. Employment opportunities have increased because of computers. Machine operators, computer technicians, data analysts, computer engineers, software engineers, programmers,

   network administrators, lab attendants and computer tutors and trainers have been recruited and this had improved the society's standards of living and thus improved the economy in general because these people pay taxes to the country.
5. It has improved data management and reduced inventory costs (costs of storage) whereby a lot of data can be stored on computer in a very organized way and a lot of space has been saved
   as we shifted from the paper files, which consumed a lot of space to electronic files
6. Computers have facilitated the availability of stronger and more reliable defense systems. This has been through the easy accessibility of information through satellites, internet and designing of sophisticated weapons, usage of CCTV cameras, finger print scanners and smoke detectors.

DISADVANTAGES

1. Moral degradation/decadence/decay. This is due to the influx of ill or immoral materials like pictures, and videos which can be accessed by everyone including the young children from the internet. This has increased the chances of rape, defilement, homosexuality etc
2. Computers have increased cases of loss of sight and other eye defects due to the radiations

   (emissions) from the computer screens. This is especially with long use of computers.
3. Consume a lot of power which has increased power bills in homes schools and other places.
4. Computers have caused a lot of unemployment in society today as a result of some tasks being done by the computer or automated. Some jobs such as messengers, typists, postmen, have reduced in importance because they can be easily done by the computer
5. Computers have increased the cost of living in our society. This is because computers require servicing and maintenance; they increase power bills, internet charges, which may be very

expensive for some people and thus stiffening the economy.

6. Computers are liable to virus attacks especially if the computers are on the internet. These viruses which are usually spread from one infected system or network usually cause heavy data loss to organizations.
7. Computers have increased the rate of forgery in society today. People use computers to forge receipts, invoices, reports and other documents to obtain money by false pretence which has led to many losses to some organizations.
8. Personal security has been jeopardized. With the internet, personal information like bank account numbers, PIN, and other personal info can be easily found by hackers who may use them in wrong deals.

Windows

Windows are rectangular work areas provided on the screen. In each window a different program or file can be run. Windows can be moved around the screen and their size and shape can be changed at will.

Icons

These are small pictures that represent commands, files, or windows. Clicking on these icons can execute a command. This makes the computer easier to use even by those who can't read.

Menus

A menu is a list of choices. GUI can allow one to execute a command by clicking at a choice from the menu.

Pointing Device

This refers to a device like the mouse which enables one to select objects by moving the pointer around the screen.

Other features include

Desktop

This is the area on the display screen where icons are grouped. Icons are meant to represent real objects on a real desk.

Pointer

This is a symbol that appears on the screen which is moved to select objects and commands. Normally, it looks like a small angled arrow though it can be changed at will.

THE COMPUTER SYSTEM

A computer system refers to the combination of components that make up a complete and functioning computer. Actually it is the co-ordination that is required between the hardware; software and human ware to enable the computer produce work or perform the required tasks.

- Let's compare the computer to the human being. A human being has got a body and a soul.

  These two parts work hand in hand for existence of a human being. In case one of the two

is not with the other, there will be no operation.

- Therefore, interconnection of the three components
- Hard ware; the tangible parts of a computer.
- Software : programs that run the computer
- Human ware
- For this case, the hardware is the body of a computer; the software is the soul of the computer, then the human ware will be the person who would like to use the machine. This person uses the software to communicate with hardware and get what he wants.

Components of a Computer System

The components of a computer system are divided into 4 categories:-

- Hardware
- Software
- Human ware/Liveware (Data Processing Personnel)
- Data

Hardware Components or Computer Hardware

- Hardware components of a computer system are the physical parts of the computer.
- They are the tangible parts of the computer. These parts can be touched and felt

What is the system unit (system case). This is a box like computer device that contains the internal components of a computer.

- There are two categories of computer hardware defined by the way they are connected to the system Unit. Those outside the system box and are connected externally are known as the ***PERIPHERAL***
- A peripheral device is any piece of hardware that can be connected to a computer. "***Peri***" means "*outside*". so they are connected the SU from outside (externally).

- Keyboard
- Mouse
- Screen/Monitor or [V.D.U.]
- Diskette
- Printer
- Loud Speakers
- Digital Camera
- Scanners
- Microphone
- Light Pen

- While those inside the system box are known as ***SYSTEM UNIT*** hardware. These can only be seen and touched when the box is opened. Such as

- Processor
- Hard Disk
- Internal speaker
- Motherboard/ Main board
- Memory
- Power supply
- Mother board

The computer hardware components are also categorized further according to the functions they perform. These are:-

1. Input devices (hardware)
2. Output devices (hardware)
3. Processing hardware/devices
4. Storage devices
5. Communication hardware.

Input Devices/Hardware

These are devices or gadgets that are used to feed data into the computer. They are devices which help to enter data into the computer.

Each input device inputs data in a specialized way. This means that each input device is designed to enter a specific type of data. So, the device to be used depends on the type of data wishes to input.

An example of data and the device that can be used is shown below:

Data Type		***Device type***
• *Graphics*	----------	*Scanner*
• *images/photos/pictures*	----------	*Digital camera*
• *sounds (Audio)*	----------	*Microphone*
• *Video (Moving pictures)*	----------	*Digital camera*
• *Text*	----------	*Keyboard*

Examples of input devices include

- Keyboard,
- mouse,
- joystick,
- scanner,
- Digital cameras,
- Touch screens.

Functions of various units and Peripheral devices of a computer.

i**) THE KEYBOARD**

This is an electro-mechanical component designed to create special standardized electronic codes when a key is pressed.

These codes are transmitted into the computer analyzed and converted into the appropriate computer usable codes. The keyboard was developed from a standard QWERTY typewriter.

The keyboard is most useful in typing texts (creating documents)

Types of keyboards

- 82 keyboard
- 102 keyboard
- 102/3 keyboard
- 105 keyboard

102/3 is most common and standard keyboard currently.
Parts of a keyboard
A standard keyboard has 3 main divisions,

- The Typewriter Area (Main Typing Area).
- The Numeric Keypad and the cursor control key.
- The Function Keys area.

ii) THE MOUSE

This is a hand held device that lets one point at the objects on the screen, click at them and move them around the screen.
The functions of the mouse include the following:

- *Right clicking*
- *Pointing*
- *Clicking*
- *Double clicking*
- *Moving the pointer*
- *Dragging*
- *Drawing pictures*
- *Scrolling*
- *Selecting/highlighting*

Clicking refers to the action of pressing the left mouse button once

Double clicking refers to the action of pressing the left mouse button twice in quick succession

Right clicking refers to pressing right mouse button.

Dragging is the act of moving items around the screen

Scrolling refers to moving the long pages up or down the screen
The mouse basically is used to move the pointer to different positions on the screen.

PARTS OF A MOUSE

The mouse has 2 or 3 push buttons that are used for its input process,

- The left mouse button or primary button
- The right mouse button or secondary button
- The wheel

The left mouse button or the primary button is the most commonly used button. It is found on the left hand side of the mouse. It carries out all the functions of the mouse apart from right clicking.

The Right Mouse button or the secondary button is found on the right hand side of the mouse. It is used for *right clicking*

The wheel is found in the middle of the left and right mouse buttons. It is used to scroll long pages up or down.

NB. The action of pressing the left mouse button once is called *clicking* and not *left clicking* as it may be the case with the Right mouse button

MICE TYPES

- *The Desktop Mouse.*

  This is commonly used with desktop PC and it has two push buttons on top and a ball on the underside. This mouse employs the principle of moving a ball which in turn moves rollers adjacent to it. These then transmit the electrical codes that relocate the pointer or cursor on the different co-ordinates of the screen.
  With the rise of the internet, a rolling wheel has been incorporated in between the two buttons. This makes scrolling the long web pages over the internet easier.

- *Track Ball Mouse (common on laptop computers).*
  A rolling ball and buttons embedded within the keyboard. Just a small head protruding through the keys on the keyboard is the one which is used to move the mouse pointer by passing a finger over it.

- *Touch pad mouse (Laptops).*

  This has a rectangular pad with 2 push buttons. The movement of a finger on the pad causes the mouse pointer to change positions accordingly.

- *The Optical mouse*

  This employs the principle of laser beam of light to transmit signals to the CPU

- *The cordless Mouse*

  This is a recent invention in the mouse technology. It does not have a cord (cable) attached to the system unit. The computer senses the mouse when it is within reach

because of the program installed in it.

iii) THE DIGITIZER

A digitizer looks like a mouse except that; it has a glass with a crosshair in the middle. The cross hair acts as a guide during the input of data. It is used in conjunction with a digitizing tablet. It is mainly used in cartography (map making) and architectural drawings to accurately trace out lines on a map.

iv) OPTICAL SCANNER

Optical scanners look like photocopiers but for them, the images are not output on paper. They are transmitted as an image into the computer. The image can be a photograph, art work or just a text.

v) JOYSTICKS AND GAME PADS

A joystick looks so much like the normal gear lever in a car. They also have the same principle like a mouse except for them they are a bit freer. They enjoy a combination of features of a mouse plus cursor control keys. This enables them rotate through a much wider angle. Joysticks are used in playing modern computer games.

vi) VOICE RECOGNITION EQUIPMENT (VRQS)

The VRQS include all types of microphones which are used to capture sound. *Microphones* allow the speaker to speak to the computer to enter data and instructions into the computer.
Musicians use multimedia computer computers to enter, edit, record (computerize) their musical tracks.

vii) TOUCH SCREENS

These are screens that can sense when a particular part of a screen is touched or pressed and hence respond accordingly. These are commonly used with security systems which can lead the fingure prints of an individual. Most PDAs use touch screens as their basic input device.
Some sophisticated mobile phones use touch screen digits instead of the press and button digits.

viii) LIGHT PENS

Light pens commonly used in high technical designs have a beam of light that is radiated into different parts of a specialized screen to input data. These beams of light are very small and sharp, therefore very precise. Light pens are commonly used in hitech designs like in the designing of integrated circuits (IC).

ix) DIGITAL CAMERA

These take photographs like the normal camera does. The major difference is that they do not create images on the film

Output Devices

These are devices which bring out the computer held information to the user. They accept/transfer information from the central processing unit and produce it in a suitable form for the user.

These devices include:-

- Monitors/VDU,
- Printers,
- plotters,
- speakers
- Projectors

Information output by these devices is either in form of hardcopy or softcopy

Hardcopy refers to the tangible copy of computer held while softcopy refers to the non-tangible (intangible) copy of computer-held information.

The common hardcopy output devices are printers and plotters while softcopy output devices are the monitors / VDU, Projectors, 3-D spectacles and loudspeakers.

1. The Monitor

This is an output device which brings out a copy of the computer held on the screen. The output in the screen is referred to as the softcopy (Non- tangible copy). The monitor works by converting information generated by the computer into visual information.

There are two main types of display devices or monitors

- cathode ray tube (CRT) or
- Liquid Crystal Display (LCD)

* A CRT- based screen, or monitor looks like the old TV set which had an extended hind part. It contains a tube which is used to project a beam of electrons at a fluorescent surface that emits light and creates images.

Advantages of CRTs

- Can produce faster
- They have a very wide viewing angle
- They are cheaper than LCDs
- They have a better resolution

Disadvantages of CRTs

- Emit higher EMR (Electromagnetic radiation) than LCD monitors
- Consume more energy than LCD Monitors

- Occupy bigger space in the room
- Not easily portable

* An LCD based screen displays visual information on a flatter and smaller screen. LCDs are frequently used in laptop computers and flat screen monitors. They use a liquid crystal rather than a cathode ray tube to create images on the screen

Advantages of LCDs

- They are lighter and therefore easy to transport
- Occupy little space on the desk or table
- Use less power
- Have little radiation and thus have less effect on the eyes

Disadvantages of LCDs

- They are more expensive than CRTs
- They have a very narrow viewing angle
- They have a relatively poor output compared to CRTs
- Slow response at times

2. Speakers

These produce sound outputs with today's multimedia programs (e.g. windows, media player), speakers have become such an important necessity. Their quality is mainly measured in Watts of power output. Most speakers use a cone diaphragm to produce sound.

4. Projectors

These are softcopy output devices. They are used to display information for bigger audiences on bigger screens or special wall. However, projectors require great light control for maximum results.

5. Plotters

A plotter is a hardcopy output. Plotters are used to produce output with thin straight and meandering lines. This is because they have a pen like print head that can accurately produce lines.

6. Printers

Printers bring out computer held information on paper or any other physical medium. Print media include paper, transparencies, cloths, some plastics, wood etc.

CATEGORIES OF PRINTERS

Printers fall into two categories. These are based on whether the image is produced by physical contact with the print media or not. The two categories are:

- *Impact printers*
- *Non-impact printers*.

1. IMPACT PRINTERS

These are printers whose print heads physically come into contact with the print medium. This is the reason why they make noise during printing.

Impact printers include:-

- Character printers
- Line printers
- Dot matrix printers.

(i) Character Printers

These are low speed printers that mimic the action of typewriters by printing one character at a time.

The characters are engraved on the print head directly. This means that the character font type cannot be changed easily. Examples include the Daisy wheel printer and thimble printer.

(ii) Line Printers

They also use the same principle during printing just like the character printers except that these are much faster. Line printers print a whole line of characters at once. They print from one end of a paper to the other end of the line hence the name.

(iii) Dot Matrix Printers

These are the most commonly used impact printers. These might either be 9-pin dot matrix printers or

24-pin dot matrix printers. Dot matrix printers form characters by the print head pins forming a pattern of dots on paper to make out a character. Dot matrix printers can also use a ribbon to create characters directly on paper.

Typical examples of Dot matrix impact printers include; the EPSON printers

Advantages of Impact Printers

1. They are not very expensive
2. Some impact printers are very fast; they can print at a speed of 160cps thus increasing productivity.

3. They are easy to maintain, their print heads only require periodic cleaning.
4. They are reliable; they can print over 100 million characters in their life span.
5. They can print on continuous paper especially dot-matrix printers

Disadvantages

1. They are noisy during operation. Because of the contact between print heads and the print medium
2. They have a lower print resolution compared to other types of printers
3. The head usually overheats during long hours of printing which slows down the process of production.
4. They are slower in printing compared to non–impact printers

2. NON-IMPACT PRINTERS

These are printers whose print heads do not come into contact with the print media when printing.

Examples of non-impact printers include:-

- *Inkjet Printers*
- *DeskJet printers.*
- *Bubble jet (Thermal inkjet) printers.*
- *Thermal printers.*
- *Laser printers.*

Inkjet/DeskJet printers

Inkjet and DeskJet printers share the same principle of printing. These work by liquid ink being spray painted onto a paper.

Liquid ink is pushed out of the printer nozzles drop by drop or in a continuous stream.

Nozzles are tiny holes in the print head through which ink passes. The ink is pushed through the nozzles by ink pumps which are attached to every nozzle.

Examples include:

Hp DeskJet 690

Epson stylus 640

Bubble Jet printers

BubbleJet printers, also known as Thermal inkjet printers. They use a bubble to force out a droplet of ink. This is done by the ink pumps heating ink in a column until when a bubble is formed. The bubble grows larger and larger until it bursts forcing out a droplet of ink through the print head nozzle on to the print media.

Since the droplet is hot it dries faster on paper than on the DeskJet.

Although these pumps work much faster, they are restricted to a rate of only 1000 dots per second because the mechanism needs time to cool otherwise the temperature would dry out the ink.

Thermal printers

These use the principle of temperature change to create characters on paper. The print heads contains dot heaters (heating elements) which cause characters to appear on temperature sensitive paper when heated.

Thermal printing papers normally discolor to a black or blue color in accordance to its make.

Advantages of using thermal printers

- They are nearly noiseless since they do not have any moving parts to generate the noise. The paper is placed into the printer and characters are formed as they get heated.
- They consume very little power (almost a sixth) of other printers.
- They are reliable because they do not experience inconveniences with paper jams or blocked nozzles
- They produce clear and crisp images with very high resolution.

Disadvantages

- They can not print on any paper because the heat generated will not cause any discoloration without burning the entire paper.
- The paper used has a limited shelf life. It can be affected by age, sunlight, humidity and other chemical vapors which may cause it to discolor completely there by causing the words to disappear.
- The paper has to be specially prepared before they can be used in printing, which creates delays.
- Their print heads can not be serviced or repaired, which makes them expensive to maintain. The whole print head must be replaced
- The printing speed is also so slow because the heads have to be allowed time to cool before the next printing cycle.
- They have a shorter life span than other printers. Their print heads are expected to print a maximum of 10 million characters accurately whereas other print heads like dot matrix printers go over 50 million characters in their life span.

Laser printers

These are high quality printers with a greater NLQ (Near Letter Quality). They have a high speed in low or no noise in printing. They can print excellent graphics.

STORAGE DEVICES OR BACKUP DEVICES

A computer can store data or information.
This can either be during or after processing.
The devices used to store data or information are called storage devices

Therefore, storage devices are components (devices) that are used in computer to record read write & preserve data or information for retrieval

Some of them can be classified as the removable data storage devices and the others as the non removable data storage devices

NB Storage devices/hardware/ *secondary storage devices* or *Auxiliary storage devices* are very essential because the memory (RAM) which is mainly used during processing as the storage space loses its contents when power is switched off.

Storage devices are divided into two categories;

- The Secondary storage devices
- The Primary Storage devices

The primary memory is the volatile memory and the secondary memory is the non volatile memory.

Basically when we talk about the data storage device, it is generally assumed to be the secondary memory because the primary storage devices for computers are actually a part of the computer's memory

SECONDARY STORAGE DEVICES

Secondary Storage devises store data in a seemingly permanent form whereas primary storage devices store data in a temporary form.

The most common secondary storage devices in micro computers are:-

- Hard disks/fixed disks
- Floppy disk/diskette
- The compact disks (CD-ROM) or CD
- Flash disks
- Tapes
- Zip drives
- Punched cards.
- Memory Card

1. Hard Disk

High-capacity storage Consists of several inflexible, circular platters that store items electronically enclosed in airtight, sealed case for protection

This is a round pan cake shaped, permanently sealed metallic magnetic storage device with in the processor. It is tightly sealed to keep out contaminants like dust and smoke.
The hard disk is also known as the **fixed disk** because it is not portable; it is always fixed within the system unit.

ADVANTAGES OF USING HARD DISKS

- The data stored on the hard disk is always secure since they reside inside the computer casing. There are fewer chances of them being stolen or misplaced.
- The destruction rate is low since they remain in one position.
- They store vast/huge/large amounts of information compared to other storage devices..
- Their speed and time of accessing data is very fast and convenient. This is because they are circular and reside inside the computer.

- Their data life is long once in use compared to other storage devices, although a few disks have been known to fail with very long periods of non-use.
- They never get misplaced or lost since they reside inside the computer.

DISADVANTAGES OF USING HARD DISKS

- Any attack of viruses causes heavy data loss. Hard disks are susceptible/prone/vulnerable to virus attacks especially if the system is unprotected.
- Hard disks expand and contract with temperature change. This may also cause data loss. This factor has led to the use of air-conditioned computer rooms. The expansion takes place because they are metallic.
- The hard disk may fail due to violent shaking and vibrations. This is why computers should never be moved about or dragged especially when they are in power.
- Transfer of data from one machine to another may be difficult if not connected to the internet or on a network. *It may require to first unscrewing opening the computer system unit (casing), which may be difficult in some cases.*
- Hard disks are very expensive to buy

2. The Floppy Diskettes

These are small portable plastic disks made of a thin circular plastic material coated with a magnetic recording surface where data is stored as tiny magnetic particles. They however store less information compared to hard disks.

Floppy diskettes are available in three basic sizes:

- 8” (8 Inch floppy diskette)
- 2.25” (5 ¼ inch floppy diskette)
- 3.5” (3 ½ inch floppy diskette)

ADVANTAGES OF USING DISKETTES

- They are portable hence the transfer of information from computer to computer is possible
- They carry enormous size of data compared to their sizes. A floppy diskette which is only 3.5” in diameter that can fit in the shirt pocket can carry a whole ream of text on paper.
- Data with a floppy diskette can easily be accessed in a shorter time.
- They are cheap to purchase.
- They are cheaper than Hard disks and CD-ROMs.
- They are portable hence used to transfer data or information from one computer to another.

They offer direct access and they are faster than magnetic tapes

DISADVANTAGES

- They are vulnerable to harsh conditions of storage like temperature changes, moisture, bending and breaking, which might cause the data life being unreliable. This is mainly because diskettes are external media and are made of plastic.

- It can lose data so quickly once the magnetic media is exposed to a magnetic field.
- They store less data compared to any other mentioned storage device. Maximum of 1.44MB of data is stored.
- They are expensive in the long run since their destruction rate is high resulting into buying all the time.
- They can be attacked by virus which causes data loss
- They have a relatively short life span
- They hold relatively small amounts of data
- They are not always 100% reliable
- They are prone to viruses

. **THE MAIN MEMORY/PRIMARY MEMORY**

It is also called Main Memory, Primary storage, Internal Memory or registers memory

Primary storage devices have direct access to the CPU .Primary storage devices store data in a temporary form helped by the register since register is a special high speed storage area within the CPU and all the data must be represented in a register before it can be processed. E.g. if 2 numbers are to be multiplied, both numbers must be in registers and the result is also placed in the register.
The register can contain the address of a memory location where data is stored rather than the actual data itself. The number of registers a CPU has and the size of each determine the power and speed of CPU. E.g. 32-bit CPU is one with in which each register is 32 bit wide. Therefore each CPU instruction can manipulate 32 bits of data.

It is classified into two:-

- Read Only Memory
- Random Access Memory

1. Ram (Random Access Memory) or RAM refers to working storage

This is the most common type of memory.

The RAM is used to keep data and programs temporary they are being worked upon. It stores programs being used on the computer at a particular time.

The RAM acts as the workspace of a computer. It can be likened to a desk or table with files in an office.

It's called Random Access Memory because it allows the user to freely read and write to it.

RAM contents are held temporarily. This is because its contents are lost once the computer or power is has been turned off. This is why we should always save our work on a secondary storage before switching off the power or turning the computer off. That's why we say its volatile memory. It is therefore advisable to keep on saving your work when using a computer

- NB.Upgrading RAM is one of the quickest means, efficient and cost effective ways to boost performance.
- Adding more RAM can add more speed to your system.

The size of RAM is very significant in determining how good the computer is. The bigger the size the better the computer's performance. Its size is measured in bytes. This is why it is regarded as the heart of the computer.

CHARACTERISTICS OF RAM

- ❖ RAM is a temporary storage.
- ❖ Its contents are user defined i.e. the user opens the programs and data to be held on RAM.
- ❖ RAM size can be changed or increased.
- ❖ The contents of RAM can be read and written to.
- ❖ RAM is volatile. This means that it can easily lose its contents
- ❖ RAM occupies the major portion of the main memory (70%).

TYPES OF RAM

There are two types of RAM:-

- Static RAM (SRAM) and
- Dynamic RAM (DRAM)

- ❖ *Static RAM* is very fast compared to DRAM and holds its contents as long as there is power.
- ❖ *DRAM* however; can hold its contents for only a short period of time even when power is on.
- ❖ The contents of DRAM can be maintained by refreshing the memory chip several times per second.
- ❖ *SRAM* is more expensive. It's used to make special types of memories like buffers and cache memory. *(Refer to the end of this topic)*

2. ROM (READ ONLY MEMORY)

After manufacturing a computer, the manufacturer design instructions that can help the computer start and also write some information about his/her computer

All these instructions and information are stored on a memory known as ROM

Therefore, Data can be stored on ROM either when the computer is being manufactured or by the computer user.

However, the disadvantages of this type of memory are that the data stored on ROM cannot be changed!!!

ROM is used to store programmed instructions and data, which are required to remain unchanged for long periods of time e.g. booting instructions and BIOS settings. This is because ROM is non-volatile; it can not lose its contents.
The contents of ROM can not be changed or deleted; they can only be read, hence the name Read Only.

TYPES OF ROM

There are four types of ROM. These depend on the permanence of the instructions or data stored there.

Mask Rom: Once data has been written there, it can never be changed.

***Programmable ROM: (PROM*)** - This allows the user to alter it only once after the content has been written on it.
Erasable Programmable ROM (EPROM) - This has a transparent quartz window through which its contents can be erased by exposing it to ultra violent (UV) light and then reprogrammed for another use.
Electrically Erasable Programmable Rom (EEPROM) This can be erased and reprogrammed using electricity. An example is the basic input and output system (BIOS).

CHARACTERISTICS OF ROM

- Its contents can only be read.
- It's non-volatile i.e. its contents can not be lost when the computer is switched off.
- Its contents are not user defined, they come from the manufacturer
- The size of ROM may not be increased or changed
- It can store instructions permanently or semi-permanently.
- It occupies the smaller portion of the main memory (30%)

This is part of the main memory which keeps only in-built set of instructions and information called ***FIRMWARE*.** The firmware is entered into the computer at the manufacturing period by the manufacturer. ROM keeps this information permanently whether power is on or off

It can store semi-permanent instructions because the latest versions of ROM chips can be programmed according to the user's specifications.

The computer can read the instructions in ROM but it can't write to it or the instructions hence the name Read Only Memory. The user can not keep his/her information in ROM.

Memory allows fast access to permanently stored data but prevents addition modification of the data.

DIFFERENCES BETWEEN RAM AND ROM

RAM	**ROM**
It is Volatile	It is non-volatile
It can be read and written to	Its contents are read only
It offers temporary storage	Offers permanent storage
Its size can be increased	Its size may not be increased
Occupies bigger portion of main memory	Occupies smaller portion of memory
Contents are user defined	Contents defined by the manufacturer

Processing Hardware

These are hardware devices that changes data into information in the computer. We have one processing device in the computer called a CENTRAL PROCESSING UNIT. Some time the call it microprocessor due to its small size.

This device is the brain of the computer and it is responsible for all operations that takes place in the computer.

The Central Processing Unit (CPU) carries out all the data processing in the computer. It's usually referred to as the brain of the computer.

The CPU is sometimes referred to as the *processor* or *central processor* or the m*icro-processor*
In terms of computing power, the CPU is the most important computing element of a computer system.
The CPU consists of three main parts:-

The two functional units and a set of workspaces that is Control Unit, Arithmetic Logic Unit (ALU) and Main Memory/Registers as its workspace.

- The control unit (CU)
- The Arithmetic Logic Unit (ALU)
- Register

NB.Some examples of the processors are Pentium 1, Pentium II, Pentium III, Pentium 4, Duo - core, i - core AMD K-6 Processors, Celeron

THE CONTROL UNIT

The control unit (or brain) controls the entire operation of the processor system during the execution of a program.

- It controls the flow of data between the main memory and the periphery devices.
- Periphery devices or peripherals are called so because they operate from the periphery (from outside) or outer edges of the system or outside the CPU and the main memory.
- The control unit determines the instruction to be executed, the task to be performed by the instruction where the results are to be stored and where the next instruction is located.
- It fetches the data from the main memory and puts it in the proper order for the processor.
- It also sends the processed results back to the main memory.
- Extracts instructions from memory decodes and executes them calling on ALU when necessary.
- Fetches and sends commands to system devices and peripherals.
- Interprets commands i.e. it's is the intermediary between the user and the computer.
- Controls and times all tasks by the CPU (typical of Multi-tasking where processor time is shared

amongst competing tasks).

- Directs the movements of electronics signals between the CPU, input, memory and output devises. Hence the CPU acting as a data traffic warden
- It coordinates all operations within the CPU (microprocessor)
- Directs data required mathematical operations from memory to ALU
- Sends processed data to either output device or storage device

ARITHMETIC LOGIC UNIT

The ALU performs arithmetic and logical operations. The arithmetic and logic computations/operations include i.e. It is responsible for all mathematical operations.

- Additions.
- Multiplication.
- Division.

It receives data from the Control Unit and performs either *Arithmetic* operations (add, subtract, multiply and divide) or *Logic* operations (making a comparison and take an action based on the result) for example *If A=2 End, Else Begin*. This example shows that if A is equal to 2 the program should end and if not equal it should start the operation again.

REGISTER is a special high speed storage area within the CPU and all the data must be represented in a register before it can be processed. E.g. if 2 numbers are to be multiplied, both numbers must be in registers and the result is also placed in the register.

The register can contain the address of a memory location where data is stored rather than the actual data itself. The number of registers a CPU has and the size of each determine the power and speed of CPU. E.g. 32-bit CPU is one with in which each register is 32 bit wide. Therefore each CPU instruction can manipulate 32 bits of data.

SPECIAL PURPOSE MEMORIES

Apart from RAM and ROM there are several types of special purpose memories found inside the CPU or in the input and output devices.

These memories are very vital because they increase the overall performance of data and instructions moving in and out of the CPU.
These memories include buffers, registers and catch memory.

BUFFERS

This is a temporary holding place that may be part of the CPU or built in an input or output device. Because the CPU is very fast compared to the input or output devices; buffers provide temporary storage so that the CPU is set free to carry out other activities instead of waiting for all data to be entered or output.

VIRTUAL MEMORY-Type of memory that supplement on the RAM space when the space of RAM is finished and the computer still has data to process,

The kind of memory where the OS allocates a portion of storage (usually on the hard disk) to function as additional RAM or the absence of RAM

CACHE MEMORY

Most modern processors incorporate small high-speed type of SRAM called Cache Memory. The purpose of cache memory is to allow the processor to access data and instructions even faster than it would have taken to fetch it from the relatively slow DRAM.

<u>HOW IS INFORMATION STORED/HELD IN THE COMPUTER</u>?

(COMPUTER MEMORY CAPACITIES)

A computer has got its own language or codes by which it understands instructions.

Two digits of 0s and ls are combined in codes of 8 digits to represent a character i.e. letters, numbers or special drawing like lines, triangles, circles, curves, etc.
Hence;

One digit (0 or 1) = a bit.
Eight digits (bits) = one byte

(Units of storage capacity)

- The standard unit for measuring computer memory is called BYTE.
- Information is kept in the computer memory in the form of binary codes (0^s and1^s).

E.g. Computer can only see 27 in a binary code as

- A single binary digit or code (i.e. 0 or 1) is called a BIT.
- A BIT is the basic unit of memory.
- BIT is an acronym (abbreviation) for **B**inary digits
- A sequence of eight bits is called a **BYTE**.
- One byte therefore consists of 8 bits e.g. 01010011.
- A single character is equivalent to one byte.
- A space between two words is also equivalent to a byte

There fore;A byte can be defined as the standard unit for measuring data or a group of eight bits put together

While A bit is as the basic unit for measuring data

There are other larger units for measuring computer data or information, which include the following:-

- Kilobytes (KB)
- Megabytes (MB)
- Gigabytes (GB)
- Terabytes (TB) this means that:

1 binary digit (0 or 1)	1 bit
8 bits	1 byte
1 kilobyte (KB)	1,000 bytes
1 megabyte (MB)	1,000,000 bytes (1000KB)
1 gigabyte (GB)	1,000,000,000 bytes (1000MB)
1 terabyte (TB)	1,000,000,000,000 bytes (1000GB)
1 Petabyte (PB)	1,000,000,000,000,000 bytes (1000TB)
1 Exabyte	1,000,000,000,000,000,000 bytes(1000PB)

How many bits are in the word PUT

- 1byte =8bits
- 3bytes=8x3 =24bits
- 4. How many bytes are in the word LADDLE?
- 1 character =1byes
- 6 =6x1
- =6bytes

COMPUTER SOFTWARE

A set of instruction or commands that tells the computer hardware what to do and how to do it

OR

Set of instructions that help the computer to process data and give information

Machine intelligence in form of written programs/instructions/procedures that cause the hardware to work

Non physical components of a computer

A set of instructions (in machine code) given to the computer to solve user problems and to control different operations of the computer.

Software is the series of instructions that tells the hardware how to work.

Computer software is sometimes referred to as the *machine's intelligence*. This is because it is the software, which directs the hardware when performing its functions.These instructions are written by someone we call **a programmer**

The act of writing/preparing theses instructions is called **programming**

The written instructions to the computer are given a name **program.**So it is very important to note that; the computer system cannot be able to process/ execute any command without software because it's the software that empowers the computer hardware. Hardware + Software = A Working Computer System

Categories of Computer Software

Computer software is sub divided into two main groups:

1. *System software.*
2. *Application software.*

The Systems software [for the computer's self operations] and Application software [for the end-user to perform tasks on the computer].Examples include: Operating systems, utility software Programming

Application software [for the end-user to perform tasks on the computer] Examples of Application software include: General Purpose also called Off-the shelf packages and these include: Word processors, spreadsheets, Databases, Desktop Publishers (DTP) and the second category known as Special purpose also called Tailor-made or customized packages has examples such as: Library Mgt system, Accounts Mgt System (for banks), Automated Teller systems, Housing/Estate management

programs .I.e. they are made basing on the need and specifications of the customer/user.

- Program: A set of instructions that command a computer to do a specific task. Examples of computer programs: Ms. Word, Ms. Excel, Scrabble game, Anti-Virus, motor arcade (game), VLC player, Skype, Nero Burning, etc

APPLICATION PROGRAM/SOFTWARE

Application software involves programs that a user can use to perform some useful tasks like writing letters, editing photos, commect to the internet, and so on

These programs are sometimes called end user programs meaning that they cannot be used to write or create other programs
The programs written by the user or professional software houses to solve specific problems

They are designed to solve user problems of a general nature (for users with a common and popular objective) e.g. producing a document, making computation, designing business posters, coming up with a database or website accounting purposes etc.
These are programs which are designed to help the user accomplish his specific tasks.

There are basically two types of application software which include:-

a. *Special purpose / Custom Designed programs*

b. *General purpose applications/Off-the-shelf programs*

a) **General Purpose/Off-Shelf Programs**

These are pre-written programs purchased off-the-shelf by the user. They are very important for non-specialists in home or business. They are not written to solve any particular problem. They are general in nature. They solve problems that are common in nature.

Advantages of General Purpose/Off-shelf packages

- Relatively low priced since they are sold in large numbers
- They are easy to use and are suitable for people with little or no computer knowledge.
- They are readily available compared to special purpose programs.
- They can be customized to solve other problems e.g. designing vouchers, CVs from template
- Have less error
- They are appropriate for a wide range of applications.
- They are usually provided together with a documentation explaining how to use them.
- Development time should be much quicker, given that most of the system will be written already.
- The software manufacturers will continually update a successful package.

Disadvantages of General Purpose Programs

- It is quite easy to forget the commands to use in the package, especially if it is not used frequently.
- Sometimes, the package will allow only a clumsy solution to the task at hand.
- The organization is purchasing a standard solution. A standard solution may not be well suited to the organization's particular needs.
- There is a high risk during implementation if the software fails to work.
- Competitors may well use the same the same package, removing any chance of using IS/IT for competitive advantage

Examples of off-shelf / General Purpose programs

- ***Word Processors*** (Ms Word, WordPerfect, WordStar, Lotus Notes, Perfect Text Editor, Professional Writer, etc). Used in creation of documents like letters, reports, labels etc.
- ***Spreadsheets*** (Ms Excel, Lotus 1-2-3, Quattro Pro. VisiCalc.) used in creation and maintenance of numerical data and carrying out calculations).
- ***Databases***, (Ms Access, dBase III and IV, MySQL, FoxBase, Paradox). They are used to keep records and files
- ***Presentation software*** (Ms PowerPoint).Used in creation of slides, slide show overheads etc
- ***Desktop publishing*** (Adobe PageMaker, Ms Publisher, Adobe InDesign) used in designing
- ***Communication software*** e.g. Browsers, internet explorer and e-mail software, Ms Outlook, Ms exchange. Used in sending mails and searching for data on the www

b).Special Purpose/Custom Made/ Custom Designed / In-House Programs

These are software programs written by users or independent software professionals to the user's specifications/requirements. This is done by purchasing the source code of the package and making modifications in house, or by paying the producer of the package to customize it

Program that is designed and written just to solve the particular problem for a particular group of people. These are not bought from shops or from the counter; they are specially written a private individual or developed in-house by someone within the organization.

A software house is a company that specializes in writing software.

Advantages of customized packages

- If the work is done in house, the organization gains considerable knowledge of how the software works and may be able to tune it so that it works more efficiently with the company's hardware.
- If done in house, the in house team may have to learn new skills.
- If done by the original manufacturer disadvantages such as those for the off- the-shelf packages may rise.

Demerits of customized packages

- It may prove costly than expected, because new versions of the standard package will also have to be customized.
- Customization mat delay delivery of the software.
- Customization may introduce bugs that do not exist in the standard version

Shareware software:

Application software that is given out to people to try it for a given time and pay for some after

FREEWARE:

Is software provided at no cost to users?

Firmware

Also as stored logic is a combination of both the software and hardware recorded permanently on electronic chips.{usually Read- Only-Memory chip that is planted into the motherboard}

Open source:

Type of application where users can be able to view the code source and make some modifications

Factors to consider when choosing a package

1. **User requirements**

Does the package fit the user's requirements?.i.e The user must be convinced of a product.

2. **Documentation**

Is there full documentation for the user? User manuals can be full of jargon and hard for a non-technical person to understand. They should be.

3. **User-friendliness**

Is the package easy to use? Is the software user friendly with the menus and clear on screen. A user-friendly package will provide prompts and will be menu-driven, giving the operator a clear choice of what to do next.

4 Market confidence/performance of the package:

The user should consider how long the package has been on the market and how satisfied the existing user is with the package

5 Technical requirement:

Where technical knowledge is required the person purchasing should ensure that the desired knowledge is possessed by the staff that will be using the package.

6 After purchase service\support:

The user should know exactly how he should be helped to solve problem if they occur after purchase

7. **Up-to-date**

How will the package be kept up to date (e.g. what if a fault is discovered in the program by software manufacturer? In an accounting package, what if the rate of VAT alters? Etc

:-

8. **Modification**

Can the package be modified by the user e.g. allowing users to insert amendments to format of the format of reports or screen displays etc? Or will the software supplier agree to write a few tailor-made amendments to the software?

9. **The cost of the package**

The firm should go for the package corresponding to the funds available. It is however important to consider the costs vs. benefits

SYSTEM SOFTWARE

These are the various programs that control the way the computer operates. They manage the operation of the computer and makes sure the computer works efficiently. System software deals with the hardware directly. System software is usually supplied with the computer hardware.
Some of these programs reside inside the ROM and are known as *firmware*.

Systems software can be grouped into the following groups

a) *The Operating system.*
b) *Utility programs.*
c) *Programming languages and Language processors*

a) Utility Programs

These are also called *service programs*. They are used to enhance the performance of the computer. The most common utility programs are:

- ***Debuggers***: - Tests the flow of a program, it detects errors in the program and notifies the user
- ***Sorting Utility***: - This allows data to be arranged as specified by the user i.e. ascending order, Descending order or otherwise.
- ***Merging Utility***: - Allows data from more than one file to be combined into one.
- ***Copying/Backup***:- This allows the creation of a duplicate copy of the original. Allows the movement of hard disk contents to backup storage.
- ***Anti-virus***:- This detects and removes viruses from the computer.
- ***Data compression***:- Helps to free space on storage device (hard disk) by removing redundant data

PROGRAMMING LANGUAGE

This is software that allows a user to write a series of instructions to define a particular task which will then be translated to a form that is understood by the computer
These programming languages help to provide a communication between the computer programmer and the machine

:-

Categories of Programming Languages

1. Low Level Languages
2. High Level Languages

Low-level Languages

These are closely associated with the processor. They include:-

- Machine Code (Object code) Languages
- Assemble Languages

High Level Languages

These were: -

- Third Generation languages (3GLs)
- They made programming much easier
- Fewer instructions were written but a lot done in less time.
- They are problem-oriented languages – made for specific operation

Third Generation Language (3GL)

There are over 500 high level languages and those which fall in the 3rd generation include;

- *PASCAL (Named after its inventor Pascal)*
- *COBOL (Common Business Oriented Language)*
- *BASIC (Beginners All Purpose Symbolic Instruction Code)*
- *FORTRAN (FORmula TRANslation)*
- *C*
- *C+*
- *C++*
- *Java*
- *Visual Basic etc.*

Fourth Generation Languages (4GLs)

They consist of pre-defined functions

With 4GL programming languages one doesn't need to know the details of the actual program codes. The codes are done automatically. The programmer only changes a few codes in order to come up with the program he wants. This has simplified programming and increased productivity.

An example of 4GL is SQL (Structured Query Language)

Fifth Generation Languages (5GLs)

They are used in intelligent knowledge based systems such as robots. They think just like human beings and are used in artificial intelligence.Unlike 4GLs, which manipulate numbers or data, these manipulate facts and rules to reach a conclusion. Therefore they can 'think', just like humans. This is why they are extensively used in Artificial intelligence projects like in Mars Exploration.

They are of two types

- High level language
- Low level language

LOW LEVEL LANGUAGES

They are called LLL because they are closely associated with the processor than HLL which are near to the programmer
That is to say; languages that can easily be understood by the computer hardware

They also of two types

- Machine code language
- Assembly code language

HIGH LEVEL LANGUGE

These are languages that are nearest to human language and can easily be understood by programmers.
They are also called English oriented Language.
They make complex programming simpler

However, it is important to note that HLL must be always first be translated to machine language before one execution
They is only one language that computes are able to understand without the use of translation program

.i.e machine language

c) The Operating System

An operating system is a group of programs that manage all the activities on the computer

This is a system program that provides an interface (connection or a link) between the computer user and the computer hardware.

FUNCTIONS OF AN OPERATING SYSTEM

The Operating System creates an environment in which the user can run other programs. The general functions of an Operating system are:-

1. Job Scheduling

The OS schedules and monitors jobs for continuous processing by the CPU. For example, it controls the loading and running of programs, communicating directly with the users and dealing with user commands to organize file and programs.

Memory Management

The operating system knows which part of the memory is in use and that one which is free. Storage space is therefore provided for data and information in the memory by the Operating system. Programs must be loaded in the memory before they are run (executed) hence the operating system has the function of choosing where they should be stored in the memory

NB: This area is called **buffer** therefore a place where data is stored 4 a short time be4 being released 4 the next processing.

a) **Processor Management**

A multi-user or multi-tasking computer system can be accessed by many users or many programs at the same time but the CPU (processor) can only run one program at a time or one user at a time. Therefore, the number of tasks or programs entering into the CPU for processing must be controlled or monitored; this is done by the operating system. The operating system uses the CU to control these processes.
Note: A Multi- user system is a system that allows many users at the same time while a Multi-tasking allows many programs to run at the same time.

Managing peripherals devices:

The operating system communicates with the peripheral device through the use of devices driver to allow data input and information output.

Peripheral device: are devices attached to the system unit from the outside environment. These devices are linked up with the OS and they use a program called **device driver**
A program that interprets data to the devices and enable installation of multiple devices at the same time

b) **Input / Output Management**

Programs required in the use of Input / Output devices and in a Multi-tasking system a conflict may arise on who is to use which device, at what time and for how long. The operating system solves this by monitoring the state of each Input an Output devices and sends a signal if any faults are detected and to indicate which device can be used at that time.

Resource Control

The operating system controls the use of computer resources by other system software and application programs being executed. These resources include input and output devices and the CPU.

2. Providing the user interface

The type of interface used on the computer's screen is provided by the operating system. An interface provides the way through which the user interacts (interfaces) with the computer.

3. Memory Management

Operating systems determines how much memory is allocated to a particular program whenever it is being used and for how long. Because the RAM is a scarce resource the operating systems determines which data remains and which one is pushed out after execution.

4. Error Handling

The operating system deal with errors produced during program execution and keeps the computer running when errors occur. It also notifies the user and normally gives suggestions on how to handle it.

:-

5. File Management

The operating system keeps track of the information in the computer, its location, use and status. The collective facilities which manage computer files are called the ***file system.***
This file system is provides a means for sorting, retrieving and sharing files. The operating system may

also provide a means of protection for data files and programs against unauthorized access and corruption.
It can also show the program being run, ensuring that users authorized to use the computer system access it especially in big systems.
Note: A file is a collection of related information. A folder is a collection of files

CLASSIFICATION/CATEGORIES OF OPERATINMG SYSTEMS

There are several ways of categorizing operating systems and these include:-

i. *According to the number of users it can support simultaneously.*
ii. *According to the number of tasks it can perform simultaneously.*
iii. *According the user interface it provides*
iv. According to the way or where they are used.

According to the way or where they are used may include.

- *Desktop Operating systems*
- *Networking Operating Systems*
- *PDA Operating systems.*

a) ACCORDING TO THE NUMBER OF USERS

- ***Single-user operating systems***

A single user or single tasking system allows one user to operate one program at a time. It cannot support more than one person and supports only one user program at a time. An example is MS-DOS.

- ***Multi-user operating system***

A multi-user computer can be accessed by several users or applications simultaneously. They can be installed on a computer that can be accessed by all users in a place. Examples include UNIX,

b) ACCORDING TO THE NUMBER OF TASKS

- ***Single tasking operating system or single program operating system***

This allows the processing of one application program in the main memory at a time. The user can only run one interactive program at a time. Before one can use another he must first quit the current running one. An example of this is Ms DOS.

- ***Multi- tasking operating systems***

This allows the computer to support more than one application program at the same time. The operating system through processor scheduling allocates time and switches from one task to another so quickly that it appears as if they are being executed simultaneously. Examples include almost all windows operating systems including Windows NT/2000, UNIX, Novell, Linux,

c) ACCORDING TO THE USER INTERFACE

Operating systems can be categorized according to the type of user interface they provide to the user. The term *user interface* refers to the interaction (interaction) between the user and the computer. (the way how a computer presents its self to a user, and how the user responds to it)The user interfaces include the following:

- *Command line interface (CLI)*
- *Graphical user interface (GUI)*
- *Menu driven interface*

- ***Command-line based operating systems***

Command-driven Operating Systems lets the user type a command at a command prompt. Commands are in form of special words or letters like Copy, print, cut and abbreviations like DEL, REN, CHKDSK etc. The commands are executed after pressing the *Enter key* or the *Return carriage*.

- ***Menu Driven Interface***

Menu driven interfaces provide the user with a list of options to choose from. The interface is suitable for beginners who may have difficulties recalling commands.

- ***Graphical User interface (GUI)***

This is a pictorial representation of tools or information on the computer screen. GUI is the latest effort to make the interface user friendlier.
GUI allows one to enter commands by pointing and clicking at pictures (icons) that appear on the screen. It allows the use of a mouse to click at icons or graphics on the screen.
GUI provides a number of features which makes it very user-friendly. These features are

:-

Icons

These are small pictures that represent commands, files, or windows. Clicking on these icons can execute a command. This makes the computer easier to use even by those who can't read.

Menus

A menu is a list of choices. GUI can allow one to execute a command by clicking at a choice from the menu.

Pointing Device

This refers to a device like the mouse which enables one to select objects by moving the pointer around the screen.

Other features include

Desktop

This is the area on the display screen where icons are grouped. Icons are meant to represent real objects on a real desk.

Pointer

This is a symbol that appears on the screen which is moved to select objects and commands. Normally, it looks like a small angled arrow though it can be changed at will.

Examples of Operating systems which provide GUI include:-

- Windows 3.x
- Windows 95
- Windows 98
- Windows NT
- Windows 2000
- Windows XP
- *Windows Vista*
- *Windows me*
- *Novell Netware*
- *Apple Mac OS*
- *Linux*

Therefore, Windows Operating Systems are types of Graphical User Interfaces (GUI).that is based on graphics instead of text and characterized by the use of Mouse and drop-down menus.

COMPARISON OF DIFFERENT OPERATING SYSTEMS

DOS (CLI)	WINDOWS (GUI)
• Requires little memory	- More memory
• Requires less disk space	- More disk space
• Not user friendly	- User friendly
• No ability to multi-tasking	- Multi-tasking ability
• Cannot be easily corrupted (destroyed) i.e. more stable	- Easy to corrupt

:-

FACTORS TO CONSIDER WHEN CHOOSING AN OPERATING SYSTEM

Qn. What factors should be considered when choosing an operating system?

- The computers specification for example memory capacity, processor speed, hard disk capacity etc.
- The type of computer in terms of size and make. This would tell you to buy either a PDA or Desktop Operating system
- The application software intended for the computer. This because some applications cannot be supported by particular operating systems
- User friendliness of the operating system
- The cost of the operating system
- Reliability and security provided by the operating system
- The number of processors and hardware it can support
- The number of users it can support

WORD PROCESSORS AND WORD PROCESSING

Introduction

A word processor is application software that enables a person to create, save, edit, format and print text documents.

Word processing therefore, is the act of using a word processor to create, save, edit, format and print text documents.

A wide variety of word processors exist on market and they include:

- Microsoft Word
- Corel WordPerfect
- Ms Works
- Word Pro.
- Lotus Notes
- WordStar.
- Lotus WordPro
- Word pad
- NotePad

Applications of word processors

- Writing letters
- Writing reports
- Projects
- Typing and creating boos
- Writing memo
- Designing Curriculum vitae etc

Some modern word processor versions can now be used to edit graphics and images and other objects.

Advantages of Using Word Processors

- A document can be stored in the computer for future reference in computer files and these files do not occupy physical space like those on paper
- Using word processors is easier and more efficient because some actions are automatic for example, the word wrap which takes the cursor to the next line if the word can not fit in the current line
- Word processors have special editing tools such as spelling and grammar checker and the thesaurus which help the user to the grammar and spelling mistakes
- It is very easy to correct omissions and errors in the work without affecting the structure and neatness of the work.
- All processors have superior formatting features

SPREADSHEET PROGRAMS

These are application programs used for the manipulation of numeric data and carrying out calculations. It is a ledger sheet that lets you enter, edit and manipulate numeric data
There are two types of spreadsheets

- *Manual Spreadsheets*
- *Electronic Spreadsheets*

Manual Spreadsheets

These are simply sheets of paper divided into rows and columns, which can be used to enter values for example; amounts of money manually using a pen or pencil. These are common in book keeping and in accounting offices.

Electronic Spreadsheets

Electronic spreadsheets are computer based. They are computer programs, which enable the user to enter values in rows and columns just the same way it is done in manual spreadsheets.
Electronic spreadsheets allow one to manipulate values (numbers) mathematically using formulae.

Advantages of Using Electronic Over Manual Spreadsheets

1. They can carry out calculations automatically which simplifies the manual and mental work of calculating which is tedious.
2. It can present numerical data on charts e.g. Pie-charts, bar graphs, line graphs etc, which makes data clearer and becomes easier to interpret.
3. Provides the user with facilities for handling databases in order to organize, process structured information.

4. It carries out a variety of tasks like financial planning, budget, and cost projections, cost planning, budgeting, producing invoices etc
5. They are speedy, efficient and flexible which makes them better tools for data/information analysis.
6. They have the aspects of speed, accuracy and efficiency which enable the user to accomplish

Disadvantages of Spreadsheets

1. It's not easy to create documents like letters, reports, or books in this program.
2. It is usually difficult to print spreadsheet because the print area must be set before printing.
3. One error created in calculating formulae can distort all the results in the worksheet.
4. Some formulae may be difficult to formulate
5. They only work when there is electricity, which means that when there is no power all work stalls

- *Ms Excel,*
- *Lotus 1-2-3,*
- *VisiCalc,*
- *SuperCalc,*
- *Ms Frameworks,*
- *VP-Planner*

PRESENTATION SOFTWARE

Presentation software refers (applications) to software applications/programmes that are used to enhance the communication process with the use of visual aids like presentation slides.

Applications used to present information using simple texts, graphics, pictures, sounds, video etc. Presentation software is very essential in effective communication because they assist the user to explain ideas graphically and dramatically.

The audience can also interpret the communicator's ideas easily (because of illustration).

Examples

- Harvard Graphics
- Lotus Freelance.
- Microsoft power point.
- Micro graphic presenter e.t.c.

Uses of Presentation Software

- Reduces boredom- interesting graphics.
- Help to present and illustrate an idea in quality and different way- photos, videos,

animations and diagrams e.g graphs.
- It can be linked to www and distributed.

Disadvantages of presentation software

- Light control might not be possible in some rooms. Sometimes its too dark and at times too much light.
- Technology may break down in the middle of the presentation.
- The teaching does not become adaptive and reactive.
- It depends on the skills and creativity of the presenter.

Microsoft PowerPoint

A graphical application that is capable of adding illustrative graphics with text in the same document. It can also include motion effects to enhance the presentation.

INTERNET TERMINOLOGIES AND INTERNET PROTOCOLS

Internet Relay Chat (IRC)

It is a multi-user live chat function where users exchange messages by typing. It enables group discussion.

Gopher

A technology that is used to make files available over the Internet. It is a menu-based method of searching for information on the Internet.

Online service

A business that provides its subscribers with a wide variety of data transmitted over telecommunication lines. Enables subscribers to communicate with one another.

Bulletin Board Systems (BBS)

This is an electronic message center. You can view messages and also leave messages if you want.

File Transfer Protocol (FTP)

A way of transferring files over the Internet from one computer to another. It is often used to download files from Internet sites.

Newsgroup (Usenet)

These are ongoing discussion groups of people on the Internet who share a mutual interest.

Email Communications

E-mail* or *Electronic mail is the transmission of messages via a computer network. It is a combination of a username and a domain name that identifies a user who sends or receives email.

Factors affecting communication speed

i) Processor speed:

A faster processor will allow surfing the web at a faster speed than a slower one.

ii) Computer Memory

A computer with more available memory will tend to surf faster. To avoid slow-downs;

- Avoid opening unnecessary software applications
- Upgrade your systems memory

iii) Capacity of Hardware.

Hubs, switches and Network Interface cards have their own maximum speeds.

iv) Heavy Internet Traffic

As traffic increases, the speed decreases. Web traffic tends to increase throughout the day and peaks around early morning.

v) Bandwidth

The maximum amount of data that a transmission media can carry at a time. The higher the bandwidth, the more data and information the channel can transmit.

A protocol is set of communication standards that control communication activity on the Internet. An IP is number assigned to any Internet-connected computer

Transmission Control Protocol (TCP)

A set of rules used to send data between computers over the Internet.

File Transfer Protocol (FTP)

A method of moving files between two Internet sites.

Hypertext Transfer Protocol (HTTP)

A set of rules for exchanging files (text, graphic images, sound, video) on the web.

Simple Mail Transfer Protocol (SMTP)

It is used to transport mails on the Internet.

Post Office Protocol (POP)

A communication technology for retrieving emails from a mail server.

Internet Services

The Internet provides the following services:

i) ***Electronic Mails (E-mails)***

the exchange of electronic letters, data, and graphics on the Internet.

ii) ***The World Wide Web (WWW)***

A vast space on the Internet where information pages are installed (websites).

iii) ***Telnet***

It allows to logon to another system while working on another.

iv) ***Chat (Chat room)***

It permits users to chat with each other by typing lines of text on the computer.

v) *Newsgroup*

An online area where users conduct written discussions about a particular subject.

vii) ***E-learning***

learning through interaction with special programs on the Internet.

viii) ***E-commerce***

A financial business transaction that occurs over the Internet.

ix) ***Video Conference***

A meeting between two or more geographically separated people who use a network or Internet to transmit audio and video data.

x) ***Internet telephony -Voice over IP (VoIP)***

A web-based telephone service that allows a user to talk to others for just the cost of the Internet connection.

COMPUTER NETWORK

A net work is the global interconnection of computers

Advantages of Networks

i) Peripherals can be shared e.g printers
ii) Software can be shared
iii) Data can be shared
iv) Computers may communicate
v) User access may be restricted.

Disadvantages of networks

i) If one server breaks down may affect a number of computers.
ii) It is vulnerable to hackers and viruses
iii) Cabling and installation may be expensive

Types of computer networks

i) ***Local Area Network (LAN):***

A network that covers a relatively small area e.g a building. Most LANs connect workstations and PCs.

The following characteristics differentiate one LAN from the other:

Topology – the physical arrangement of devices.

Protocols - the rules and encoding specifications for sending data

Media – the cables that connect the devices e.g Twisted-pair wire, coaxial or fiber optic cables.

ii) ***Wide Area Network (WAN):***

A network that covers a large geographical area. WANs consist of two or more LANs connected through public networks such as Telephone lines, satellites, etc. the largest WAN is the Internet.

iii) ***Metropolitan Area Network (MAN):***

A network that covers a relatively small geographical area such a Town, city, etc.

Bandwidth

The maximum amount of data that a transmission media can carry at a time. The higher the bandwidth, the more data and information the channel can transmit from one computer to another.

Requirements to access the Internet

i) *Computers*

ii) *Network Interface Card (NIC)* – it connects the computer to the network.

iii) *Network hubs and cables* – a hub connects multiple computers at a central location.

iv) *MODEM* – a link between analog signals and digital signal of a computer.

v) ***Internet Service Provider (ISP)***

This is a company that provides Internet services. They include:-

- UTL – Uganda Telecom Limited
- MTN – Mobile Telephone Network
- AFSAT, etc

Factors in choosing an ISP

- Reasonable pricing
- Faster Internet performance
- Provides help and installation
- Security for its clients
- Can communicate with you

Internet Service Provider (ISP)

An ISP is a company that enables companies or individuals to connect to the Internet. ISPs always provide:-

- Email capabilities
- Web Hosting
- Connectivity, etc.

ISP include: MTN , UTL, Africaonline, Infocom, etc.
Online Service Provider (OSP)
These are companies that provide proprietary content and services accessible only to their members. Members use their own proprietary software to access their services. Such services include:

- News
- Sports
- Weather
- Online chat groups, etc.

Example is Americaonline.

COMPUTER VIRUSES

A computer virus is a computer program/code that copies (can copy) itself and infects a computer without the knowledge of the user.

A computer program/code specifically designed to damage or cause irregular behavior in other programs on a computer. It is designed to infect and affect the computer's performance negatively.

Types of Viruses

There are many software codes regarded as computer viruses that can cause damage to computer systems. They include:-

- *Boot sector viruses*
- *File viruses*
- *Hoax viruses*
- *Trojan horses*
- *Backdoors*
- Boot sector viruses

- *Worms*
- *Jokes*
- *Test viruses*
- *Time Bombs*
- *Multipartite viruses*

This destroys the booting information on the computer or storage devices. It affects the booting files and causes failure in booting.

File viruses

These attach themselves to computer files. They usually delete files or cause erratic behaviors in the file system.
Hoax viruses

They are usually lies; they don't exist in real sense. They only exist in the imagination of the press and public. They don't exist despite the rumor of their distribution and creation. The rumour about their creation and spread is usually spread through e-mails or the press.

Trojans / Trojan horse

This is a small program code hidden within legitimate software. The software continues to work normally until such a time when they are activated to cause trouble. A Trojan horse does not have the capacity to replicate (copy) itself like a worm.

Worms

This sticks in the computer memory and re-writes (replicate) itself in the memory until it can't multiply anymore. This causes the computer to stop working because the memory is full.

Backdoors

This may be a Trojan or a worm that allows hidden access to a computer system.

Droppers

These are programs that have been written to perform useful tasks like compressing files, previewing video clips etc. but end up introducing viruses in the system in the process of performing their functions.

Failed virus

These are viruses that have failed to meet their goals. This is mainly due to poor programming by the author/designer or is intercepted by a real virus. They are common on the internet when downloading software.

Packagers

These hide the existence of a virus from virus guard by masking some codes around the actual software programs. It is only when a virus has appeared that you realize the software had a virus.

Test virus

These are simply written to test some virus guard or anti-virus software. They are not harmful, just for learning purposes only.

Jokes

A joke is a harmless program that does amusing things on the screen. E.g. Messages like-"*Your computer is about to explode in five minutes, please run away……."* These messages appear and disappear in few seconds.

Sources of Viruses

Viruses spread in various ways but the most common ways are:-

Fake Games

Computers games are a common source of viruses because most games are irresistible. Virus designers design fake games and attach virus onto them. Once a game is executed, a virus is run and activated. These games keep infecting systems as they are installed

Contaminated systems

Contaminated computer systems can spread virus if used freely. For example, installation diskettes for a particular application program can introduce viruses on to a system whenever that application is installed. There by spreading the virus. It is also common for pirated software.

Freeware and shareware

These soft wares are usually given/downloaded free of charge. These are good grounds for distributing viruses.

Freeware are software programs which are usually distributed on line free of charge while *shareware* refers to those software programs, which can be shared freely amongst the users.

Legitimate Software Updates

Software may get a virus from software house during programming by unscrupulous virus authors or during the distribution say through a network, where they get viruses from the wild viruses across the internet.

Pirated software

The use of pirated software introduces the risk that the software may be contaminated by virus code or amended to perform some other destructive function which may affect the system. Pirated software is that which was copied illegally with an aim of making profit

Ways of Spreading Viruses

Viruses are commonly spread or activated in 3 basic ways:

- Opening an infected file
- Running an infected program
- Starting up the computer with an infected floppy diskette
- Use of infected storage devices like floppy diskettes, hard disk etc.
- Through E-mails or distributed maliciously through the internet.
- Through downloads from the internet especially free ones
- Through freeware and shareware.

Symptoms of Viruses

- Unfamiliar graphics or quizzical messages appearing on screens.
- Programs taking longer than usual to load.
- Disk accesses seeming excessive for simple tasks
- Unusual error messages occurring more frequently
- Less memory available than usual
- Programs and files disappearing mysteriously.
- Computer indicating that the storage devices are full.

Disasters Caused By Virus

- Damaging programs/software
- Deleting files/data on storage devices
- Formatting the hard disk.
- Boot failure
- Take up / fill up the computer memory
- Causes system crashes.
- Corruption of files
- Slows down the speed of the computer

Precautions Taken Against Viruses (Control Measures)

- Ensure that there are regulations and a policy on the usage of computers and their protection (e.g. no foreign diskettes unless first scanned)
- Ensure that the e-mails are from a trusted source before opening them or e-mail attachments
- Avoid opening e-mails before scanning them for viruses
- Install anti-virus utility and update its virus definitions frequently for detecting and removing viruses.
- Never start up a PC with a floppy diskette in the drive.
- Scan all the drives and files for possible virus infection before using them.
- Write protect the recovery disk before using it.
- Back up important files regularly.

Anti-Virus Utility/ Program

An anti-virus utility is a program that prevents, detects and removes viruses from a computer's memory or storage devices. This utility is installed into the computer by the computer user or owner to perform the very purpose identified above. The user is required to check or clean the computer regularly using

this anti-virus. Examples of common anti-virus programs *include Kaspersky, AVG, Norton, and MacAfee, Avila, Avast* etc

DATABASE MANAGEMENT SOFTWARE

A database is a collection of related data about a particular subject or for a specific purpose. A database is a collection of structured and related data items organized so as to provide a consistent and controlled access to the items.

Examples of database programs include;

1. Microsoft Access (Ms- Access) Preferred for small desktop databases
2. Oracle- preferred for larger corporate databases
3. FoxPro
4. Database IV
5. Lotus Approach
6. Microsoft Visual FoxPro
7. Borland database
8. Corel Paradox

In DBMS one can:-

- Sort data
- Create reports
- Create forms
- Organize queries
- Make tables e.t.c

There are mainly two types of databases;

- *Manual databases*
- *Electronic databases.*

However, Electronic databases are today commonly used.

Manual databases

These are non-computerized. These are traditional ways of recording and keeping data using simple devices e.g. a book with a list of items in stock, a diary e.t.c.
However, these methods/databases (manual) had /have problems/weaknesses.

- Poor update of records
- Time wasting (when searching for particular item).
- Unnecessary duplication of data
- Misleading reports due to poor entry and organization.

Electronic databases

These are computer based databases. They are further categorized into Distributed database, which sit on individual stand-alone computers and centralized databases, which sit on a database server.

Advantages of electronic databases over manual

- Enhancement of data integrity because it is centralized, fewer updating errors occur and greater accuracy can be maintained.
- Ensured data independence because data is entered, stored, modified and accessed by methods that are not affected by application programs.
- Reduction of data redundancy – storing data in one place means less duplication and less space is required for storage
- Facilitates data sharing and integration – database systems offer users the ability to combine or to cross reference data in many different ways.
- Improvement of access to data – data systems allow users to query the data base directly without necessarily using application programs
- Centralization of security – it is easier to limit access to information it it is grouped together instead of being kept in several scattered files. Most databases must be protected and kept private.
- Reduction of costs – data entry, data storage, and the development of new application programs are all made more economical by eliminating the duplication of data, organizations can realize substantial savings.

Disadvantages of Electronic Databases

- Difficult to develop
- Expensive to manage.
- Require highly-trained expertise for maximum management.
- Data security may not be ensured in case of data sharing.
- Data can be pirated or corrupted by unscrupulous data managers